

Kia & Hyundai Theft Data Story

Roshan GC

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Audience

This story is written for City of Milwaukee public-safety decision makers—MPD command staff, the Mayor’s office, and theft-prevention partners (community organizations and insurers). They understand operational metrics (volume, peaks, staffing) but are not expected to interpret raw tables or complex statistical jargon. The visuals therefore prioritize clear comparisons, time patterns, and locations that support immediate decisions.

Purpose and Call to Action

The intent is to convert four Kia/Hyundai theft datasets into an actionable prevention plan. **Call to action:** adopt a coordinated “Kia/Hyundai Theft Reduction Package” that combines (1) accelerated outreach for immobilizer/software updates and steering-wheel lock distribution, (2) targeted patrol/surveillance deployment during peak months and repeat locations, and (3) public messaging that reduces opportunistic theft (lock use, parked-vehicle guidance, rapid reporting).

Medium

A short, slide-based narrative (six figures) is used here as a Word report to meet the assignment requirement and to keep the story linear and decision-focused.

Data and Visuals (R)

Instructor note addressed: the six figures below are **non-duplicative**—each answers a distinct question (timing, composition, concentration, annual trend, geography, and cross-city benchmark).

```

packages <- c("tidyverse","lubridate","readxl","janitor","scales","zoo","maps","treemapify")
installed <- rownames(installed.packages())
to_install <- setdiff(packages, installed)
if(length(to_install) > 0) install.packages(to_install)

library(tidyverse); library(lubridate); library(readxl); library(janitor)
library(scales); library(zoo); library(maps); library(treemapify)

```

```

mke <- readr::read_csv("/users/roshan/DSC640/KiaHyundaiMilwaukeeData.csv", show_col_types = FALSE) %>%
  mutate(month_year = my(paste(month, year))) %>% arrange(month_year)

multi <- readr::read_csv("kiaHyundaiThefts.csv", show_col_types = FALSE) %>%
  mutate(month_year = my(paste(month, year))) %>% arrange(city, state, month_year)

mapdf <- readr::read_csv("carTheftsMap.csv", show_col_types = FALSE)

vice_raw <- readxl::read_excel("Motherboard VICE News Kia Hyundai Theft Data.xlsx", sheet = "Data", col_names = TRUE)

head(mke)

```

m...	y...	city	state	countKiaHyundaiThefts	countOtherThefts	percentKiaHyundai
<chr>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
Jan	2019	Milwaukee	WI	22	235	0.086
Feb	2019	Milwaukee	WI	13	218	0.056
Mar	2019	Milwaukee	WI	10	195	0.049
Apr	2019	Milwaukee	WI	10	238	0.040
May	2019	Milwaukee	WI	11	280	0.038
Jun	2019	Milwaukee	WI	15	330	0.043

6 rows

```
vice_colnames <- names(vice_raw)
subheader <- vice_raw[1, -1] %>% unlist(use.names = FALSE) %>% as.character()

city_names <- vice_colnames[-1]
city_names <- ifelse(str_detect(city_names, "^Unnamed"), NA_character_, city_names)
city_names <- zoo::na.locf(city_names, na.rm = FALSE)

vice_data <- vice_raw[-1, ] %>%
  rename(date = 1) %>%
  mutate(date = as.Date(date))

lookup <- tibble(
  col = vice_colnames[-1],
  city = city_names,
  metric = subheader
)

vice_tidy <- vice_data %>%
  pivot_longer(-date, names_to = "col", values_to = "value") %>%
  left_join(lookup, by = "col") %>%
  select(date, city, metric, value) %>%
  mutate(
    metric = case_when(
      str_detect(metric, "Kia") ~ "kia_hyundai",
      str_detect(metric, "All") ~ "all_thefts",
      str_detect(metric, "Percent") ~ "percent",
      TRUE ~ as.character(metric)
    ),
    value = as.numeric(value)
  ) %>%
  filter(!is.na(city)) %>%
  pivot_wider(names_from = metric, values_from = value) %>%
  mutate(percent = ifelse(percent > 1, percent/100, percent))
```

Figure 1 (Stacked area): When does Milwaukee

experience theft pressure?

```
fig1_df <- mke %>%
  select(month_year, countKiaHyundaiThefts, countOtherThefts) %>%
  pivot_longer(-month_year, names_to = "type", values_to = "thefts") %>%
  mutate(type = recode(type,
                        countKiaHyundaiThefts = "Kia/Hyundai",
                        countOtherThefts = "Other vehicles"))

ggplot(fig1_df, aes(x = month_year, y = thefts, fill = type)) +
  geom_area(alpha = 0.9) +
  scale_x_date(date_breaks = "6 months", date_labels = "%Y-%m") +
  labs(
    title = "Milwaukee vehicle theft volume is rising, with Kia/Hyundai thefts contributing increasingly",
    x = "Month", y = "Thefts (count)", fill = ""
  ) +
  theme_minimal(base_size = 11) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

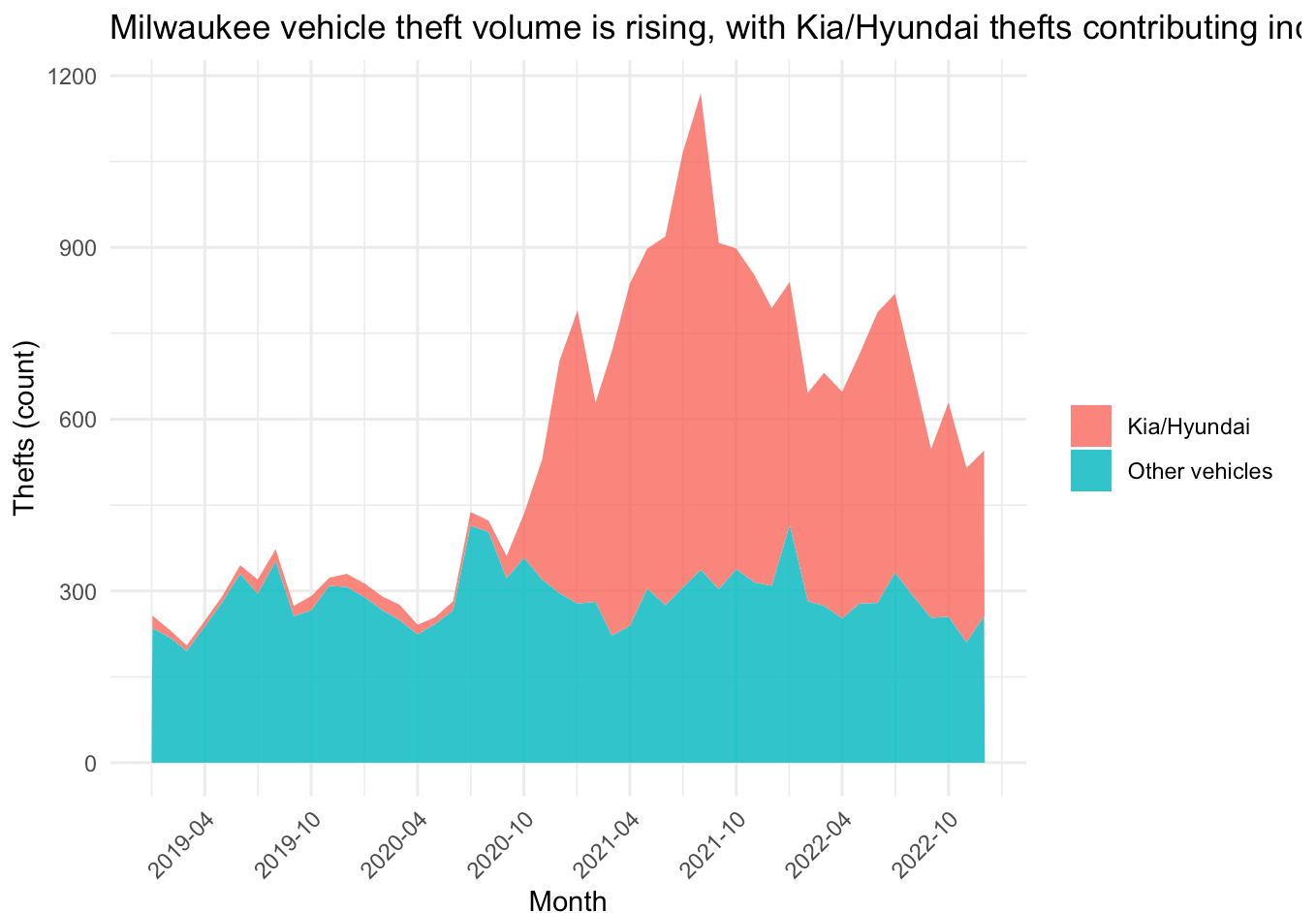


Figure 2 (Donut): How large is the Kia/Hyundai

share in 2022?

```
fig2_df <- mke %>%
  filter(year == 2022) %>%
  summarise(
    `Kia/Hyundai` = sum(countKiaHyundaiThefts, na.rm = TRUE),
    `Other vehicles` = sum(countOtherThefts, na.rm = TRUE)
  ) %>%
  pivot_longer(everything(), names_to = "type", values_to = "thefts") %>%
  mutate(pct = thefts/sum(thefts))

ggplot(fig2_df, aes(x = 2, y = thefts, fill = type)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  xlim(0.5, 2.5) +
  geom_text(aes(label = percent(pct, accuracy = 0.1)),
            position = position_stack(vjust = 0.5), size = 4) +
  labs(
    title = "In 2022, Kia/Hyundai thefts represent a substantial share of Milwaukee vehicle thefts",
    x = NULL, y = NULL, fill = ""
  ) +
  theme_void(base_size = 11) +
  theme(legend.position = "bottom")
```

In 2022, Kia/Hyundai thefts represent a substantial share of Milwaukee

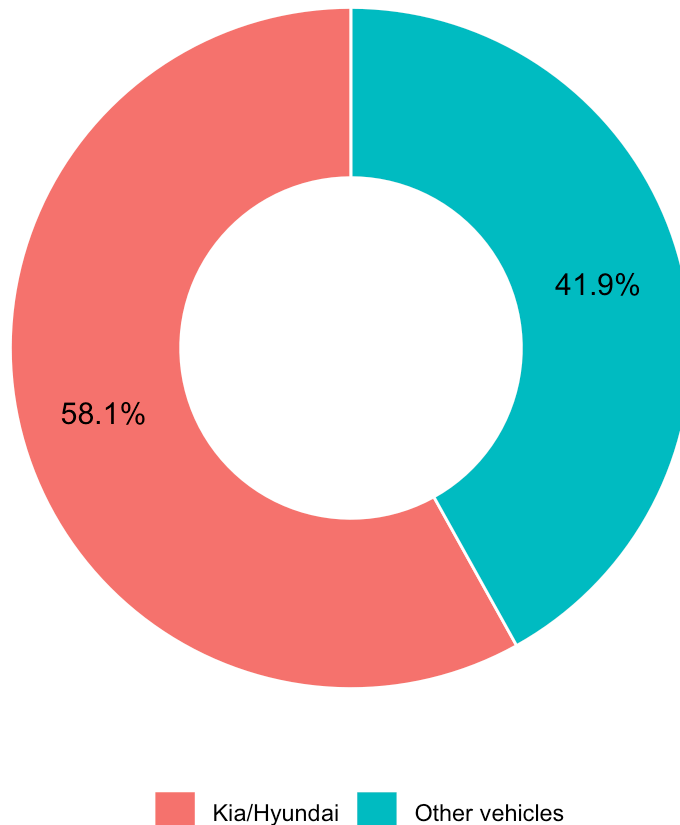


Figure 3 (Treemap): Where are Kia/Hyundai thefts most concentrated nationally?

```
fig3_df <- multi %>%
  group_by(city, state) %>%
  summarise(kia_hyundai_thefts = sum(countKiaHyundaiThefts, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(kia_hyundai_thefts)) %>%
  slice_head(n = 12) %>%
  mutate(city_state = paste0(city, ", ", state))

ggplot(fig3_df, aes(area = kia_hyundai_thefts, fill = city_state, label = city_state)) +
  geom_treemap() +
  geom_treemap_text(reflow = TRUE, place = "centre", min.size = 9) +
  labs(
    title = "Kia/Hyundai thefts are concentrated: top cities account for a large share of incidents (2019–2022 total)",
    fill = ""
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "none")
```

Kia/Hyundai thefts are concentrated: top cities account for a large share of incidents (:

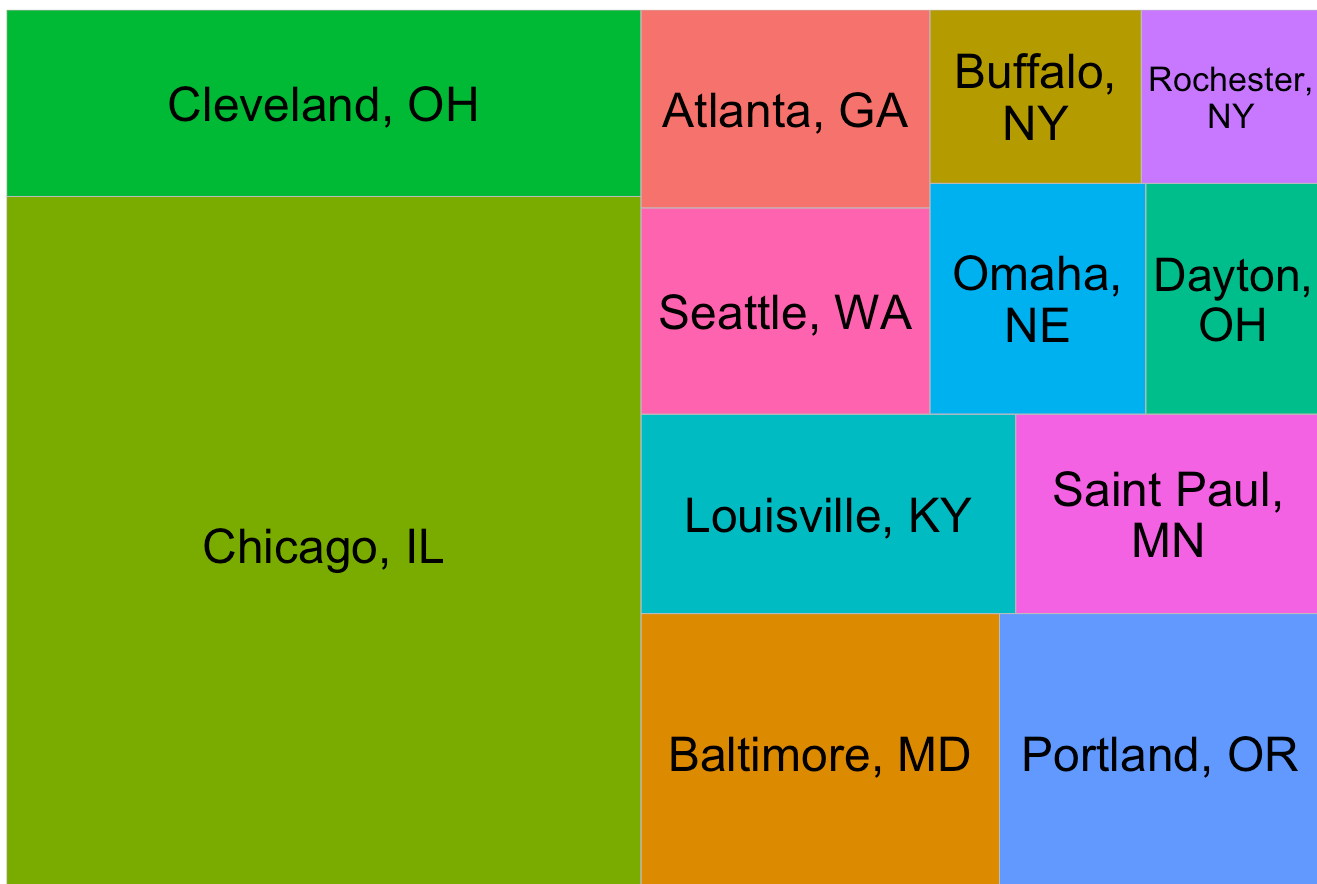


Figure 4 (Stacked bar): How is the problem changing year-over-year in Milwaukee?

```
fig4_df <- mke %>%
  group_by(year) %>%
  summarise(
    `Kia/Hyundai` = sum(countKiaHyundaiThefts, na.rm = TRUE),
    `Other vehicles` = sum(countOtherThefts, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(-year, names_to = "type", values_to = "thefts")

ggplot(fig4_df, aes(x = factor(year), y = thefts, fill = type)) +
  geom_col() +
  labs(
    title = "Year-over-year, Milwaukee's vehicle theft growth includes a growing Kia/Hyundai component",
    x = "Year", y = "Thefts (count)", fill = ""
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom")
```

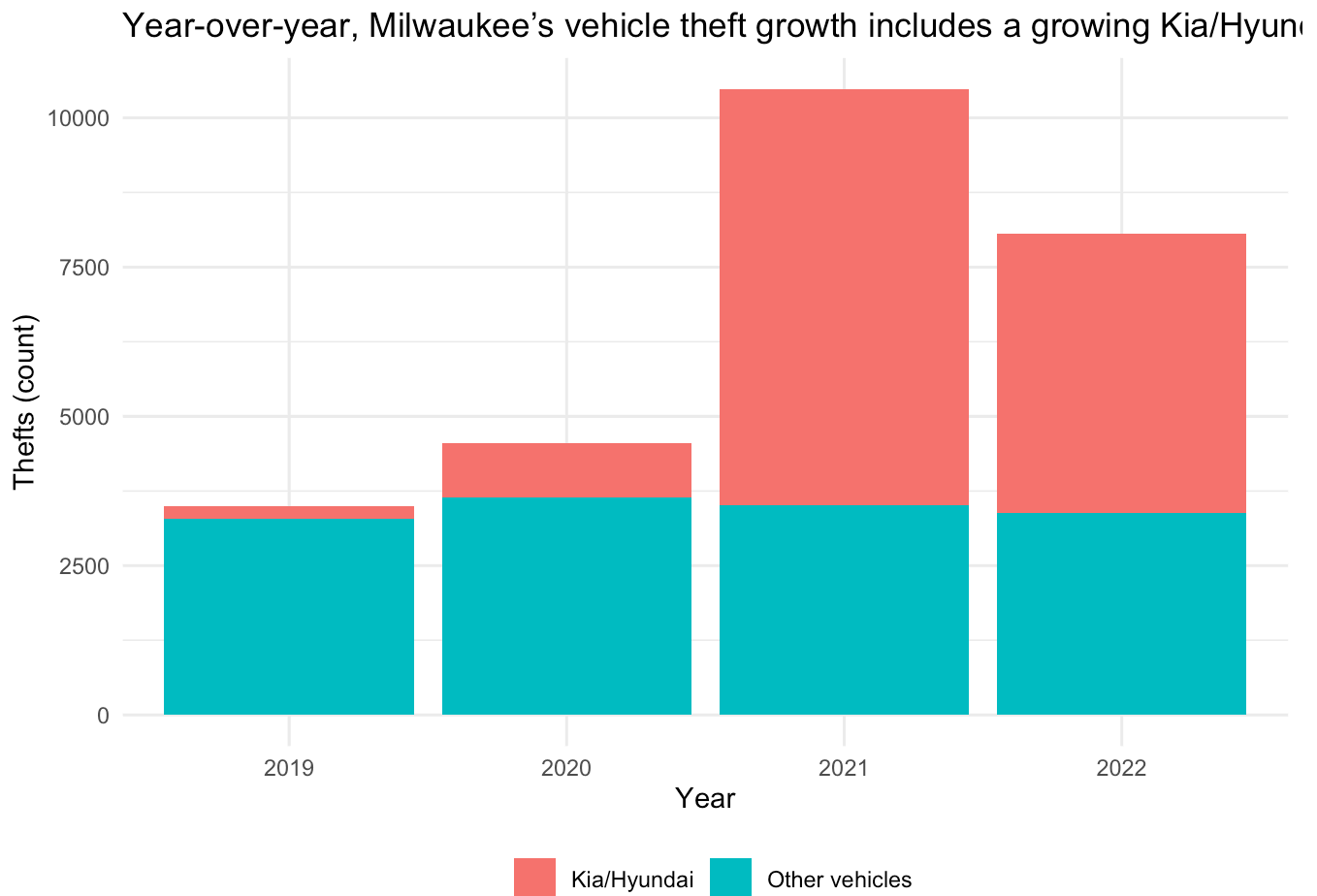


Figure 5 (Map): Which agencies show the biggest change from 2019 to 2022?

```
us_states <- map_data("state")
map_plot_df <- mapdf %>%
  mutate(change = percentChange2019to2022,
         change_dir = if_else(change >= 0, "Increase", "Decrease"))

ggplot() +
  geom_polygon(data = us_states, aes(x = long, y = lat, group = group),
             fill = "grey95", color = "white", linewidth = 0.2) +
  geom_point(data = map_plot_df,
            aes(x = longitude, y = latitude, size = abs(change), shape = change_dir),
            alpha = 0.6) +
  coord_fixed(1.3) +
  scale_size_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    title = "Many agencies show large changes in total vehicle thefts from 2019 to 2022",
    x = NULL, y = NULL, size = "|% change|", shape = ""
  ) +
  theme_void(base_size = 11) +
  theme(legend.position = "bottom")
```

Many agencies show large changes in total vehicle thefts from 2019 to 2022

•

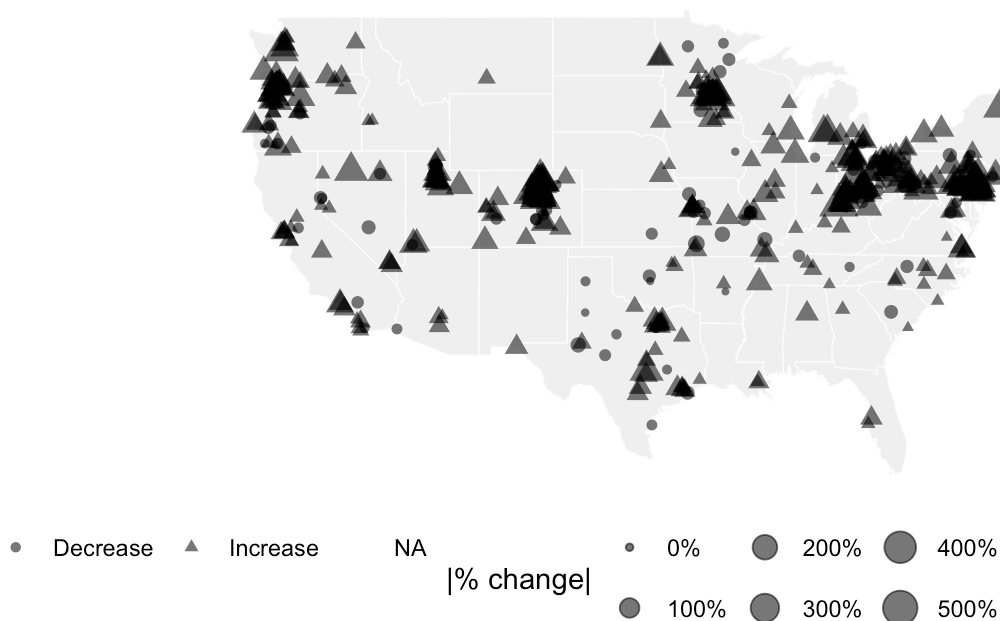


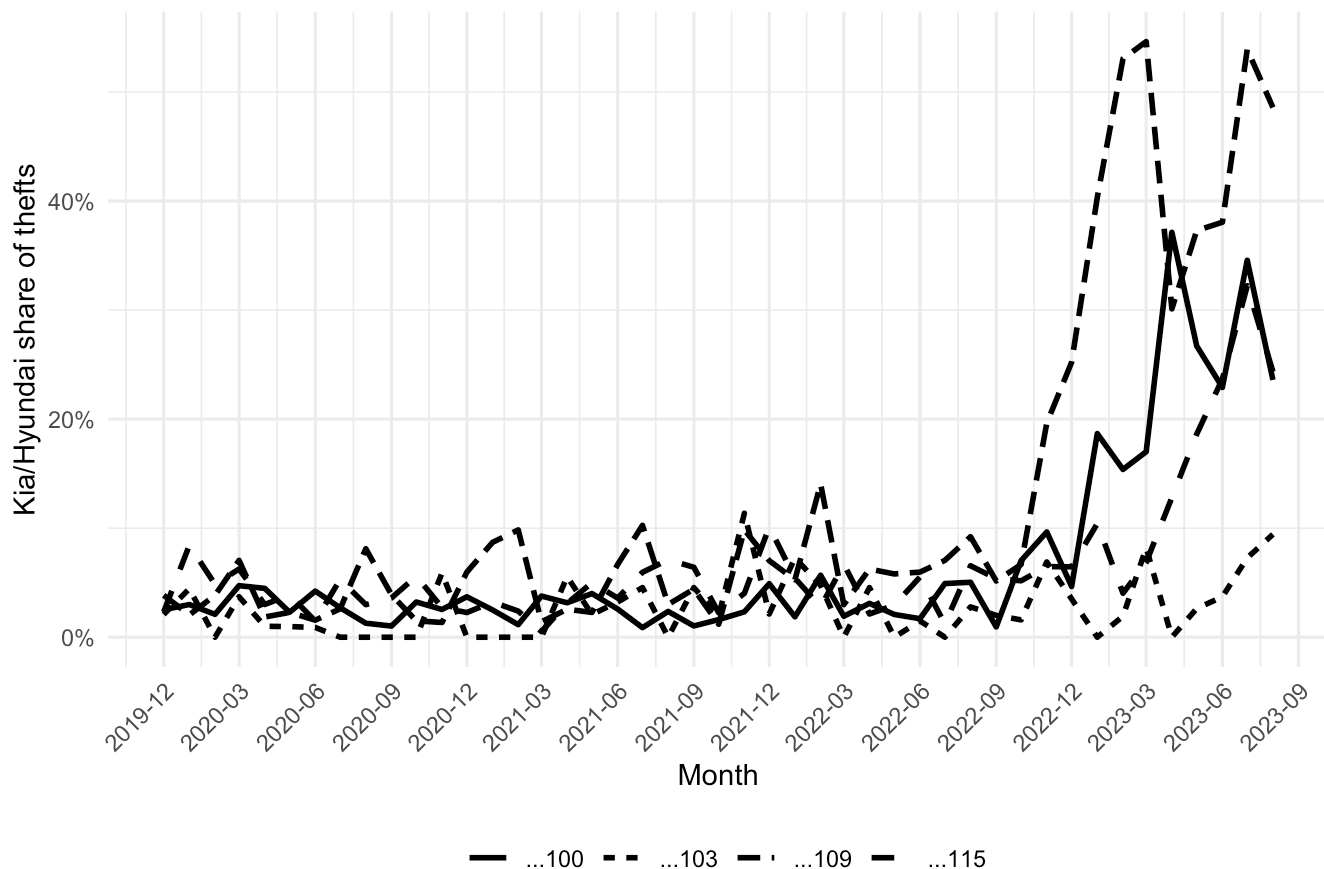
Figure 6 (VICE benchmark): How does Kia/Hyundai share vary across cities?

```
top_cities <- vice_tidy %>% filter(!is.na(percent)) %>% count(city, sort = TRUE) %>% slice_head(n = 4) %>% pull(city)

fig6_df <- vice_tidy %>% filter(city %in% top_cities, !is.na(percent)) %>% arrange(city, date)

ggplot(fig6_df, aes(x = date, y = percent, group = city, linetype = city)) +
  geom_line(linewidth = 1) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  scale_x_date(date_breaks = "3 months", date_labels = "%Y-%m") +
  labs(
    title = "The share of thefts that are Kia/Hyundai varies across cities (VICE dataset)",
    x = "Month", y = "Kia/Hyundai share of thefts", linetype = ""
  ) +
  theme_minimal(base_size = 11) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "bottom")
```

The share of thefts that are Kia/Hyundai varies across cities (VICE dataset)



Design Choices

Design choices follow Gestalt principles to reduce cognitive load. Similar colors represent Kia/Hyundai thefts consistently across figures, while stacked views use proximity and common region to signal “parts of a whole.” Alignment and shared scales enable fast comparison across months and years. Titles are written as conclusions (not labels), and annotations highlight peak months and highest-change geographies so the audience can act quickly.

Ethical Considerations

Key transformations include standardizing dates (month/year → a single month index), aggregating totals (monthly → annual, and multi-city totals), and filtering to top cities or agencies for readability. These steps risk hiding smaller areas; mitigation is to label filters clearly and provide the full data tables in an appendix. The map uses aggregated agency counts and percent change, avoiding individual-level identifiers. Because crime data can stigmatize neighborhoods, locations are presented at the agency/geo-name level and framed as prevention opportunities. Sources are documented from the provided course datasets and linked articles; any missing values or inconsistent city naming are reported in a data notes section.

Findings and Conclusion

Milwaukee’s time series shows seasonal surges and a strong upward shift in Kia/Hyundai theft counts in later years, indicating predictable operational pressure. Composition views show Kia/Hyundai thefts forming a growing share of overall vehicle thefts. National agency patterns reveal large 2019–2022 increases in multiple jurisdictions, suggesting a broader vulnerability rather than a single-city anomaly. Cross-city benchmarking (VICE dataset) shows that the Kia/Hyundai share varies widely, supporting targeted interventions where the share is highest. Overall, the evidence supports prioritizing update/lock distribution and surge staffing during peak periods while tracking progress monthly.